because between measurements, nothing matters. Even talking about the existence of things between measurements is unscientific. This, strangely enough, is the orthodox view of quantum physics—the “soft pillow” of the Copenhagen interpretation.

These suspiciously easy answers raise another question, one without an obvious solution. Physics is the science of the material world. And quantum theory purports to be the physics governing the most fundamental constituents of that world. Yet the Copenhagen interpretation says that it’s meaningless to ask about what’s actually going on in quantum physics. So what is real? Copenhagen’s reply is silence—and a look of stern disapproval for having the temerity to ask the question in the first place.

This is, at best, a profoundly unsatisfying answer. But this is also the standard answer. The physicists who pursued the question anyhow—physicists like Einstein, and later on, Bell and Bohm—did so in open defiance of Copenhagen. So the quest for reality is also the story of that rebellion, a rebellion as old as quantum physics itself.